

Multiple Choice

1. **Answer:** B

Options: A) Physical layer; B) Application layer; C) Data link layer; D) Network layer

Note: A proxy firewall operates at the application layer because it intercepts and analyzes the traffic between a client and server, allowing it to filter content based on application-specific protocols and data.

2. **Answer:** B

Options: A) Confidentiality; B) Availability; C) Correctness; D) Integrity

Note: Availability is not considered a classic security property like confidentiality and integrity, as it focuses on ensuring access to information rather than the protection of that information itself.

3. **Answer:** C

Options: A) Eavesdropping; B) Cross-site scripting; C) Authentication; D) SQL Injection

Note: Authentication is a fundamental security process used to verify the identity of a user or system, rather than a security exploit, which typically involves malicious actions aimed at compromising security.

4. **Answer:** A

Options: A) $1/K$; B) $K-1/K$; C) $\log_{10}(1/N)$; D) $N-1/m$

Note: In a complete K-ary tree of depth N, the total number of nodes is given by $(K^{(N+1)} - 1) / (K - 1)$, while the number of nonterminal nodes, which are all internal nodes, is $K^N - 1$, leading to the ratio of nonterminal nodes to total nodes being approximately $1/K$ as N becomes large.

5. **Answer:** B

Options: A) Memory leakage; B) Buffer-overflow; C) Less processing power; D) Inefficient programming

Note: A buffer-overflow occurs when a program writes more data to a buffer than it can hold, leading to potential overwriting of adjacent memory, which can be exploited by attackers to gain unauthorized access or disrupt system functionality.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Wireless security encompasses protocols, technologies, and practices designed to safeguard data integrity and prevent unauthorized access to devices connected through wireless networks.

7. **Answer:** False

Options: A) True; B) False

Note: Church's thesis, which posits that any effectively calculable function can be computed by a Turing machine, remains unprovable as it is a foundational assumption in computer science rather than a theorem that can be definitively proven or disproven with advancements in computational theory.

8. **Answer:** False

Options: A) True; B) False

Note: WPA 2 is not a recent protocol; it was established in 2004 as an enhancement to WPA and is widely used for securing wireless networks, rather than being a new feature that emerged in recent years.

9. **Answer:** True

Options: A) True; B) False

Note: The set of Boolean operators {AND, OR} is not complete because it lacks the ability to express negation, which is essential for representing all possible Boolean expressions.

10. **Answer:** False

Options: A) True; B) False

Note: A front-door typically refers to a legitimate access point in a system, but it may not always be fully traceable or compliant with security protocols, as vulnerabilities can exist even in authorized access methods.

Long Form Response

11. **Answer:** `def alternate_elements(list 1): | return result`

Note: correct because it outlines the essential function structure needed to generate a new list containing alternate elements from the provided list, indicating an understanding of list manipulation and function definition in programming.

12. **Answer:** `def sort_mixed_list(mixed_list): | return int_part + str_part`

Note: The answer correctly outlines a function that separates integers and strings from a mixed list and returns them concatenated, demonstrating an understanding of data types and list manipulation in programming.

End of Answer Key